



Research article



Carbon capture through 3D bioprinting and in situ biomineralization: investigation of physical and mechanical properties

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ABSTRACT

Global warming has fostered the development of new technologies to convert CO₂ to valuable products. Inspired by natural mineralization mechanisms, this study presents a novel approach to capture CO₂ through microbial mineralization in 3D-printed bioconstructs. The proposed approach employs biocatalytic carbon capture facilitated via carbonic anhydrase enzyme produced by *Sporosarcina pasteurii* bacteria. Therefore, the developed living bactoink is capable of capturing CO₂ by in situ biomineral trapping (i.e., transforming CO₂ to CaCO₃ in the printed constructs). For this purpose, the optimized blend of carboxymethyl cellulose and alginate is first obtained to prepare the required hydrogel basis of bactoink. Next, the bactoink comprising bacteria and hydrogel is prepared, and its printability is assessed. Finally, a comprehensive evaluation of the physical properties and mechanical performance of the printed constructs is conducted after their exposure to CO₂. The CaCO₃ precipitation is confirmed by SEM, XRD, and Raman spectroscopy. The method shows rapid strength gain of the printed bio-constructs, achieving up to 8.6 MPa after 2 days of mineralization, highlighting promising characteristics for real-world applications. These findings underscore the potential of biocatalytic 3D-printing as a viable strategy for sustainable carbon capture and utilization. Furthermore, unlike common microbially induced carbonate precipitation (MICP) techniques, our proposed method relies on direct microbial CO₂ capture, effectively reducing undesirable by-products such as ammonium ions produced in conventional ureolytic MICP. 3D bioprinting of living CO₂ absorbing constructs not only facilitates the synthesis of bioinspired materials for industrial and biomedical applications but also advances construction technology for earth and extraterrestrial settlements.

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1. Introduction

Global CO₂ emission has surpassed 41 billion tonnes in 2024 [1]. The rise in the atmospheric levels of carbon dioxide (CO₂), which serves a notable role in accelerating global warming, is one of the most important environmental challenges of our time [2,3]. The increase in CO₂ levels has prompted significant concerns among scientists and policymakers. The surge in CO₂ emissions has resulted in various adverse environmental effects, including ocean acidification, forest fires, and glaciers melting, thus posing a significant threat to Earth ecosystems [4]. To combat this critical environmental issue, it is essential to develop effective, sustainable, and durable carbon dioxide mitigation strategies.

Nature always presents efficient and eco-friendly solutions for engineering problems. Biological utilization of CO₂ naturally occurs in biosystems such as bacteria, plants, and algae [5]. Throughout the lifespan of the Earth, microorganisms have controlled the fate of carbon in land and marine environments [6]. Therefore, bio-based carbon capture strategies, which harness the power of microorganisms, present a natural candidate and sustainable substitute for conventional techniques. Conventional carbon capture and utilization (CCU) techniques often result in undesirable byproducts, high energy requirements, and pollution [7,8]. A significant advantage of microbial techniques for CO₂ capture over chemical procedures is their ease of growth and absence of toxic byproducts. The shortcomings of existing catalytic approaches, such as limited efficiency, harsh conditions, and high running costs, underscore the need for bio-solutions to CO₂ sequestration. Metabolic pathways of microorganisms can be employed to transform CO₂ into beneficial organic compounds like glucose and carboxylic acids, paving the way for cutting-edge carbon capture and storage technologies that support carbon neutrality and sustainability [9]. In addition, carbon has naturally been captured through conversion to inorganic materials (i.e., biomineralization mediated by microorganisms has been crucial to the Earth carbon cycle in marine ecosystems and oceans).

Biominerals are biogenic materials produced by many living organisms, including vertebrates. Biomineralization has played a pivotal role in the physiological function of various living organisms. In vertebrates, biominerals form organs and tissues essential for locomotion (e.g., bones), food mastication (e.g., dentin and enamel), hearing capability, and gravity sensation (e.g., otoconia) [10]. In plants, biominerals (i.e., the so-called cystoliths) scatter light into the light-deprived center of leaves [11]. Furthermore, they have been essential in plants and animals' protection against herbivores and predators (e.g., mollusk shells and spines of sea urchins) [12, 13]. By taking different forms and functions, these biocrystals have evolved across the Earth's history, enabling different eukaryotes and prokaryotes to sense and adapt to their environmental conditions [14]. In addition to the multifunctionality of biomineral tissues, their exceptional mechanical properties have attracted scientists. The durability of naturally cemented cliffs of Western Australia formed by Stromatolites throughout millions of years, and the mechanical resistance of mollusk shells have encouraged studies to develop bio-inspired and functional biomaterials for industrial and medical applications.

While synthetic materials are often limited by their intricacy in terms of composition, structure, and function, biomaterials and biomimetically-generated materials favor precise control over composition and structure. Their biosynthesis offers new perspectives for producing high performance and environmentally friendly materials [13,15]. Inspired by the structure of the shell and spines of sea urchins, a biocomposite formed of proteins and biominerals, new materials such as elastic concrete, capable of enduring fractures, have been invented [16]. Mimicking chemical, structural, and mechanical properties of natural biocomposites has been a source of inspiration for developing biocomposites used in industrial and biomedical applications (e.g., filters, membranes, and scaffolds) [17].

Microorganism-mediated mineralization provides an energy-efficient process to develop biocomposites. Biomineralized light-weight (yet mechanically strong) materials have been proposed for skeletal repair [17]. Bioinspired hydrogels, locally stiffened by minerals, have been employed to repair bone cracks [18]. Nanocrystals of hydroxyapatite produced by *Bacillus tequilensis* render suitable candidates for tissue engineering and bone healing [19]. Incorporating living microorganisms in biocomposites enables the production of biocomposites with dynamically evolving characteristics and composition, enabling better control of their structural complexity and function. Living biocomposites can also possess self-repair and self-healing characteristics [20]. Furthermore, it allows the exploitation of the microbial metabolic pathways for processes of interest.

The microbial metabolic processes often result in favorable environmental conditions for precipitating various minerals such as silicates, carbonates, phosphates, and ferrates [21]. Microbially induced calcium carbonate precipitation (MICP) has been an emerging technique to produce low CO₂ footprint materials. Production of one ton of cement results in approximately 1 ton of CO₂ emission [22]. In a quest to find eco-friendly alternatives to cement, MICP has been employed to enhance the mechanical properties of construction materials [7], produce self-healing concrete [23], stabilize soils [24] and plug cracks. In the MICP technique, negatively charged microbial cells attract calcium cations, which then bind with carbonate ions, present in the medium, to produce calcium carbonate [24]. Therefore, bacteria offer two important roles for biomineralization. Their cells provide the required nucleation sites for mineralization, and the enzymes involved in their metabolic processes facilitate the production of carbonate ions. In urease producing microbial strains, hydrolysis of urea can be utilized for carbonate generation, and thereby calcium carbonate precipitation (Equations (1)–(4)).



While ureolytic MICP relies on the urease enzyme and urea hydrolysis to produce low carbon emission biocement, it is preferred to directly use CO₂ to sequester carbon, achieving negative emission materials. In order to achieve CO₂ capture through mineralization, it is required to increase the solubility of carbon dioxide. The slow kinetics of CO₂ hydration are an innate barrier (rate constant, $k \sim 0.04 \text{ s}^{-1}$) for industrial scale CO₂ capture [25]. To facilitate photosynthesis and respiration, nature expedites the solubility of CO₂ by the carbonic anhydrase (CA) enzyme, present in various plants, animals, and microorganisms [26]. CA enzyme with a rate constant as high as 10^6 s^{-1} can greatly enhance CO₂ solubility [27]. CA enzyme is encoded in the genome of bacterial strains such as *Sporosarcina pasteurii* [28,29]. Carbonate ions produced by the hydration of CO₂ react with available calcium cations in the medium, resulting in biologically mediated CaCO₃ precipitation. Therefore, CA-based biomineralization can effectively be employed for CO₂ capture as described by Equations (5) and (6).



Additive manufacturing through 3D printers has revolutionized various fields such as construction technology, drug delivery, tissue engineering, and biosensors [30]. It allows the fabrication of customized and complex structures of different shapes while maintaining minimal material waste. In order to form biomimetic structures, 3D bioprinters have recently emerged, which utilize bioinks prepared from natural hydrogels to fabricate biomaterials. Bio-sourced hydrogels (biopolymers) are not only biocompatible but also suitable for generating light-weight high strength biocomposites [31]. Cell-laden bioink has been another recent advancement, enabling the synthesis of functional living materials [32]. Integration of biomineralizing bacterial strains with additive manufacturing techniques provides new opportunities for construction technology and biomedical engineering [15]. The available studies integrating bioprinting and MICP rely on hydrolysis of urea for biomineralization, which results in ammonium ions, bearing toxicity and bringing environmental concerns. Alternatively, direct exposure of a printed bactoink (comprising CA-producing bacterial strains embedded in hydrogels) to CO₂ is the most eco-friendly approach to fabricate and bioprint negative emission materials, which has not yet been investigated to the authors' knowledge.

Recently, developing self-healing enzymatic materials (by direct incorporation of CA-enzyme in cementitious paste [33]), as well as CO₂ absorbing enzymatic biofilters (by constructing an absorption column containing CA-enzyme [3]), have been proposed. These studies relied on the use of CA cell-free enzymes, which need to be extracted from plants and microorganisms. The extraction process is often costly, and the enzyme stability and durability are restricted by environmental conditions. Furthermore, the lack of bacterial cells limits the mineralization efficiency of enzymatically induced carbonate precipitation (EICP), due to the lack of nucleation sites for mineral precipitation [23]. These challenges are not faced by the direct use of bacterial strains producing CA-enzyme in the bactoink. Furthermore, as compared to the absorption process, CO₂ capture via biomineral trapping (inside 3D printed constructs) allows for direct CO₂ utilization and synthesis of living CO₂ absorptive materials.

In order to address these research gaps and limitations, a new approach has been proposed in this study that synergistically combines biocatalysis with advanced 3D printing technologies. Particularly, we propose a urea-free bactoink to fabricate CO₂-absorbing constructs via 3D bioprinting. Inspired by natural biocomposites, a combination of hydrogels (alginate and carboxyl methylcellulose) and in situ biomineralization by CA-producing bacteria (i.e., *Sporosarcina pasteurii*) is considered to achieve high strength CO₂ absorbing living materials. By direct exposure to CO₂ injection, not only is carbon capture capability of bacterial strains utilized, but also the need for urea is eliminated, thereby reducing the risk of contaminants in the printing process. Additionally, a comprehensive evaluation of the printability, physical properties, and mechanical performance of the resulting bactoink was conducted. After CO₂ exposure, bioprinted living constructs achieved up to 8.6 MPa strength, adequate to resemble bone tissue (e.g., human trabecular bone [34]) and constructional materials (to repair structural elements and to strengthen soil as a stabilizing agent [22]). Therefore, the CO₂ exposed-printed sections can be employed in medical and industrial applications, providing innovative applications for CO₂ capture and utilization through 3D bioprinting of living organisms.

2. Materials and methods

2.1. Materials

Sporosarcina pasteurii (strain designation PTCC 1645), bacteria medium culture Nutrient Broth (Sigma-Aldrich®), potassium alginate (Kimica Corp. CAS: 6005-36-1), carboxymethyl cellulose (CMC) (Henzakchemie, Food-grade), calcium chloride (Merck, CAS:10035-04-8), urea (Sigma-Aldrich, catalog number 51456), ethanol (Merck, 99.8% purity), hydrochloric acid (Kimia Exir, 37% purity), and carbon dioxide gas were utilized without any modifications.

2.2. Preparation of bacteria

The culture medium required for bacterial cultivation was prepared by adding 25 g of nutrient broth to deionized water and preparing 1 L of solution. The solution was autoclaved for 30 min at 121 °C at a pressure of 15 psi. Inoculation of the strain was performed under a laminar flow hood. The preparation was done by inoculating a suitable growth medium with a single colony of the bacteria. The culture was incubated at a shaker incubator at a rotating speed of 200 rpm and a temperature of 30 °C for 24 h. The concentration of bacteria in the culture medium was monitored by measuring the optical density (OD) at a wavelength of 600 nm until

the bacteria reached the desired OD of 1.

2.3. Preparation of bactoink

To determine the optimal formulation for the ink used in printing applications, various ratios of bactoink components were tested. The optimized formulation consisted of 7 g of potassium alginate, 7 g of carboxymethyl cellulose (CMC), and 100 mL of the bacterial suspension (OD = 1). These components were mechanically mixed at a specified speed for 20 min. The hydrogel basis of the considered bioink was, thus, composed of both CMC and alginate. As revealed in previous studies, cellulose-based components offer greater strength and durability, whereas alginate enhances the stability and biological compatibility of the hydrogel [35]. Additionally, CMC creates a more favorable environment for the growth of *S. pasteurii* [36].

2.4. 3D printing of bactoink

Bactoink was loaded into a 5 mL Luer-Lock syringe and centrifuged at 3800 rpm for 10 min at 20 °C (HETTICH/ROTOFIX 32A) to remove trapped air bubbles. The 3D printing process utilized a modified, optimized 3D bioprinter that was in-house designed. Bactoink was extruded through a 21 G needle, powered by a screw-driven operating at a flow rate of 0.63 mL/min, with a printing speed of 28 mm/s. The distance from the needle tip to the substrate was maintained at 1 mm. The printed samples, measuring 20 mm × 20 mm × 8 mm with a grid infill pattern, were subsequently gelled in a 0.5 M CaCl₂ solution for 30 min.

2.5. Biomineralization of bactoink

For biomineralization, printed samples were immersed in a sealed container with 0.5 M CaCl₂ aqueous solution. A pressure controller maintained the container at 1 atm by purging CO₂ gas. The biomineralization process continued for up to 2 days, with samples removed for analysis every 24 h. After completion, samples were treated with ethanol to fix and deactivate bacterial activity and subsequently dried in an oven at 60 °C for 24 h. This protocol ensured termination of microbial metabolism and stabilization of the mineralized constructs for further characterization. The schematic of the bactoink preparation and printing process is shown in Fig. 1.

2.6. Rheology of bactoink

Rheological measurements were performed using a rheometer (Anton Paar-MCR 302) at a controlled temperature of 25 °C, analyzing the samples with and without bacteria to assess their rheological properties.

2.7. SEM imaging of biomineralized samples

Scanning Electron Microscopy (SEM) imaging was performed using a TESCAN Vega 3 (Czech Republic) equipped with Energy

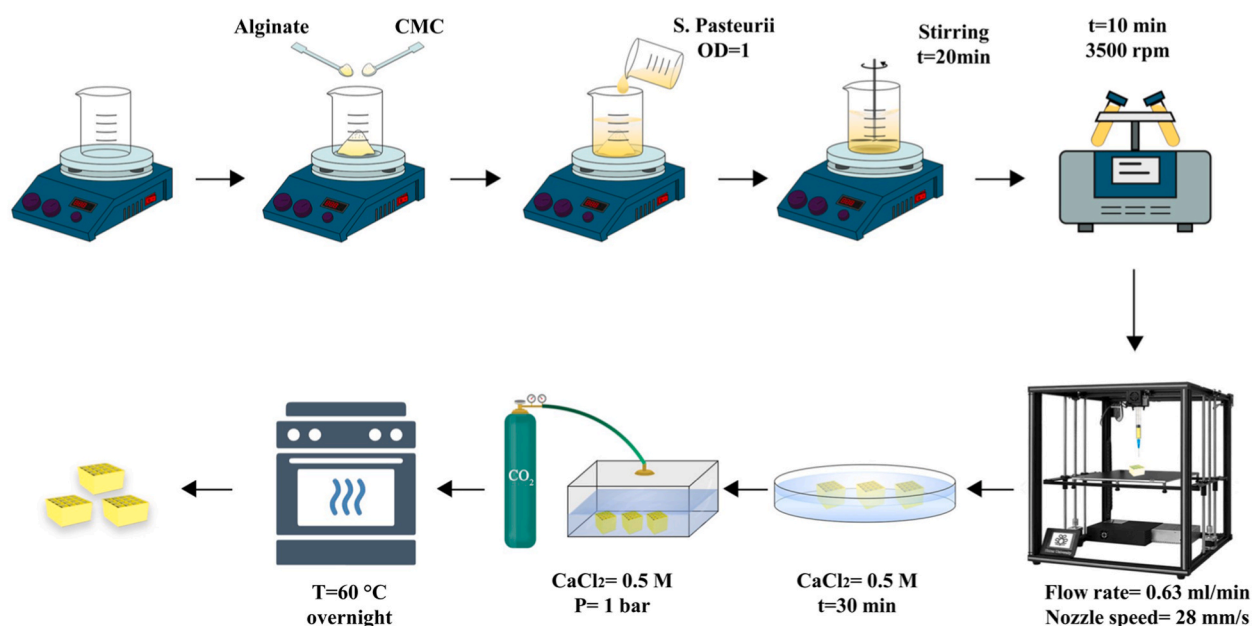


Fig. 1. Schematic of bactoink preparation and 3D printing process.

Dispersive Spectroscopy (EDS). Prior to imaging, samples were coated with a 5 nm layer of gold.

2.8. TGA measurement of biomineralized samples

Thermogravimetric analysis (TGA) of biomineralized samples was performed using a METTLER TOLEDO instrument. The analysis spanned a temperature range of 30 °C–950 °C, employing a heating rate of 10 °C/min in an inert nitrogen atmosphere. The weight percentage of calcium carbonate (wt% CaCO₃) was determined using the following equation (eq. (7)) [20]:

$$\text{wt}\%_{\text{CaCO}_3} = \frac{\Delta \text{wt}\%}{m_{w,\text{CO}_2}} m_{w,\text{CaCO}_3} \quad (7)$$

where $\Delta \text{wt}\%$ denotes the weight loss between 600 °C and 900 °C, m_{w,CO_2} is the molecular weight of carbon dioxide (44.102 g mol⁻¹), and m_{w,CaCO_3} indicates the molecular weight of calcium carbonate (100.086 g mol⁻¹).

2.9. XRD measurement of biomineralized samples

X-ray diffraction (XRD) analysis of powdered biomineralized samples was conducted using a D8 Advance-Bruker, equipped with Cu K α radiation ($\lambda = 1.542 \text{ \AA}$). The data were collected over a 2θ range of 10°–80°. Baseline removal and peak identification were achieved by the Peak Analyzer function in OriginPro 2021. The mean crystallite size was estimated using the Debye-Scherrer equation (eq. (8)).

$$D = \frac{K\lambda}{\beta \cos \theta} \quad (8)$$

Where K stands for the Scherrer constant, λ represents X-ray beam wavelength, β attributes to the half peak width (FWHM, in radian), and θ is the peak diffraction angle.

2.10. Raman spectroscopy

Raman spectroscopy was conducted using a Lab Ram HR confocal Raman spectrometer (Japan) to characterize the crystal structures of powdered biomineralized samples, employing a 523 nm laser wavelength.

2.11. Mechanical characterization of biomineralized samples

To investigate the impact of biomineralization on compressive strength, samples with a height-to-diameter (h/d) ratio of 0.5 were fabricated to mitigate buckling during compression testing. Compressive tests were performed using an AllroundLine 2005 machine (5 kN load cell, Zwick Roell). Samples were subjected to compression at a constant velocity of 0.5 mm/min until a strain of 40% was reached.

2.12. X-ray micro-computed tomography (X-ray μ CT) imaging

To investigate the changes in internal porosity of the samples, particularly inspecting any variation in porosity associated with micropores, micro-computed tomography imaging of the samples using LOTUS-NDT scanner (Behin Negareh) was conducted. μ CT imaging parameters were set as 60 kv, 100 μ A, 2000 projections (1 frame per projection) and the frame exposure time of 1 s. μ CT images were obtained at a resolution of 7.2 μ m. Image visualization and porosity determination were performed using BoneJ module and ImageJ software.

2.13. Quantification of CO₂ sequestration capacity

To evaluate the CO₂ sequestration capacity of the bioprinted specimens, quantification of biomineralized calcium carbonate content (CaCO₃%) is essential. Acid dissolution is a well-established and reliable gravimetric method for this purpose [37]. Each biomineralized sample was first weighed to obtain its initial dry mass (m_1), then individually immersed in 20 mL of 1 M HCl for 1 h to ensure complete dissolution of CaCO₃ throughout the sample matrix. This duration was selected to allow sufficient diffusion of HCl into the bulk of the samples and complete reaction with mineralized carbonate phases.

Following acid treatment, samples were rinsed with distilled water to remove residual acid and reaction byproducts. To ensure complete removal of HCl trapped within the sample structure, each sample was subsequently immersed in 100 mL of distilled water for 1 h with gentle agitation. Samples were then dried overnight at 100 °C and reweighed to determine the final dry mass (m_2). The CaCO₃ content (wt.%) was calculated as follows:

$$\text{CaCO}_3\% = \frac{m_1 - m_2}{m_1} \times 100 \quad (9)$$

All measurements were performed in triplicate to ensure reproducibility, and average values are reported. To determine the CO₂ sequestration capacity, the mass of CO₂ chemically incorporated into the precipitated CaCO₃ was calculated based on the

stoichiometry of the mineralization reactions (Equations (5) and (6)). These reactions indicate a 1:1 M ratio between sequestered CO₂ and formed CaCO₃.

Given that the CaCO₃ content is expressed as a weight percentage (CaCO₃%) of the initial dry biomineralized sample mass (m_1), the mass of CaCO₃ in the sample is $\text{CaCO}_3\% \times m_1$, while the mass of the non-carbonate matrix is $(100 - \text{CaCO}_3\%) \times m_1$. Using the 1:1 M relationship and the molar mass of CaCO₃ ($M_w = 100.09 \text{ g mol}^{-1}$), the CO₂ sequestration capacity was normalized to the mass of the non-mineralized matrix and expressed in millimoles of CO₂ per gram of matrix (mmol/g). This yields the following expression:

$$\text{sequestration capacity} \left(\frac{\text{mmol CO}_2}{\text{g matrix}} \right) = \frac{1000 \times \text{CaCO}_3\%}{M_w \times (100 - \text{CaCO}_3\%)} \quad (10)$$

3. Results and discussion

3.1. Biocalcification validation

An experimental setup was designed to validate the concept of transforming CO₂ to calcium carbonate directly using the carbonic anhydrase (CA) enzymes produced by bacteria. For this purpose, a culture medium containing *S. pasteurii* bacteria was prepared and placed in a sealed container. As discussed earlier, the calcium carbonate production test utilizing *S. pasteurii* occurs through two distinct mechanisms: urea was added to one container, while carbon dioxide was injected into another for 15 min. A schematic representation of this experimental process is provided in Fig. 2a and b. The obtained SEM micrographs confirm the presence of biogenically produced calcium carbonate, attached to bacteria in the precipitated powder (Fig. 2). This observation confirms that *S. pasteurii* can effectively produce calcium carbonate by increasing atmospheric CO₂ hydration, and subsequently its biomineralization, even in the absence of added urea. These results underscore the intrinsic capacity of selected bacteria to engage in biocalcification and CO₂ biomineral trapping processes, highlighting their potential to produce living CO₂ absorbing materials for diverse applications in environmental remediation, construction technology, industrial biotechnology, and biomedical engineering.

3.2. Mechanism of CaCO₃ precipitation via CA-based pathway of bacteria

The formation of CaCO₃ in the presence of the carbonic anhydrase (CA) enzyme from the *S. pasteurii* bacteria involves a multifaceted mechanism facilitated by the hydration of CO₂ under constant pressure conditions (Fig. 2a). While CO₂ is injected into an aqueous solution containing CaCl₂, the carbonic anhydrase mediates conversion of the aqueous CO₂ into bicarbonate (HCO₃⁻) and protons (H⁺), according to Eq. (5). This reaction increases the concentration of carbonate ions (CO₃²⁻) through the dissociation of bicarbonate, eventually leading to CaCO₃ precipitation (Eq. (6)). The CA enzyme concentration and the ratio of [Ca²⁺]:[CO₃²⁻] govern the precipitation process and thereby crystal morphology and polymorphs [28]. The type of CaCO₃ polymorphs produced in the 3D bioprinted constructs exposed to CO₂, in this study, is discussed in detail in the subsequent sections.

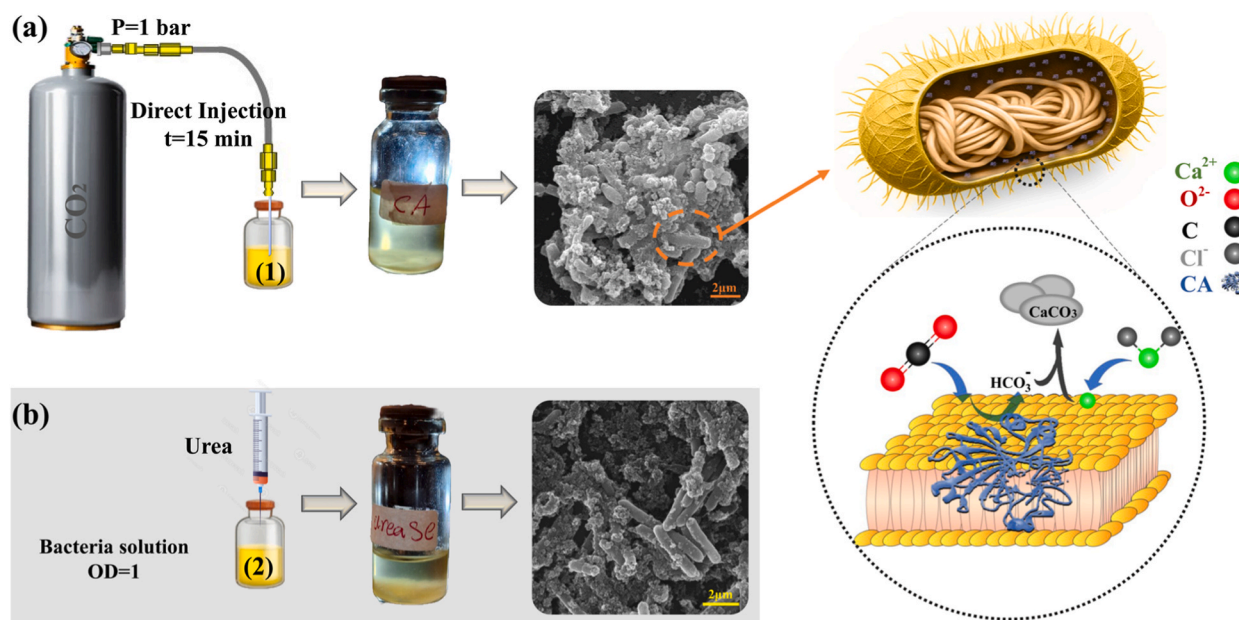


Fig. 2. A schematic representation of the experimental setup for calcium carbonate production using *S. pasteurii* bacteria and a visual depiction of the carbonate precipitates alongside the bacteria: a) Direct injection of CO₂ and its biomineralization (CA-based) pathway, b) ureolytic (urease-based) pathway.

3.3. Bactoink properties

3.3.1. Rheological characterization

Hydrogel composition has substantial implications for mechanical properties and overall performance in 3D printed applications. Achieving optimal printability necessitates a careful balance of material properties, especially viscosity. Therefore, to optimize the printability of our bioink (consisting of Carboxymethyl Cellulose (CMC) and alginate), different ingredient ratios had to be examined. According to Fig. 3a, alginate alone exhibits poor printability. To enhance its printability, CMC was added as a thickener. Being a strong thickener, even small amounts of CMC significantly improved the rheological properties of the bioink. As demonstrated, by increasing the concentrations of alginate and, particularly, CMC, printability was improved. Fig. 3a illustrates three distinct regions, transitioning from a non-printable state to a gel-like state with enhanced printability zone. This progression underscores the importance of optimizing the alginate-to-CMC ratio to achieve rheological properties suitable for 3D printing. A 7:7 alginate-to-CMC ratio was selected as the optimal formulation. This ratio ensures a sufficient amount of CMC to create a gel-like, printable bioink while providing sufficient alginate to confer high mechanical strength after crosslinking, thus assuring the structural integrity of the printed constructs.

Investigating the rheological properties of hydrogels is crucial for understanding their potential applications in bioprinting. Key rheological characteristics, particularly viscosity, significantly influence printing speed, resolution, and the overall fidelity of printed structures [38]. In addition, bioinks generally exhibit shear-thinning behavior, which is suitable for extrusion-based bioprinting, facilitating improved flow and precise control during the printing process [39].

For this reason, the viscosity test was performed on prepared samples (Fig. 3b). For further and more precise investigation of printability, the rheological properties of optimized bioink from simple extrusion tests were measured with and without bacteria. The optimized CMC: alginate ratio showed shear-thinning behavior in both conditions, which is consistent with previous observations on shear thinning of CMC [40] and alginate [41]. Due to the shear thinning behavior of CMC and alginate, similarly, their mixed solutions exhibited shear thinning properties. However, the bactoink displayed slightly sharper shear thinning in the middle range of shear rates. To quantify the degree of shear thinning of bioink and bactoink, the power law (Ostwald-de Waele) model was fitted to the measured viscosities. The model is presented in equation (10).

$$\eta = K\dot{\gamma}^{n-1} \quad (10)$$

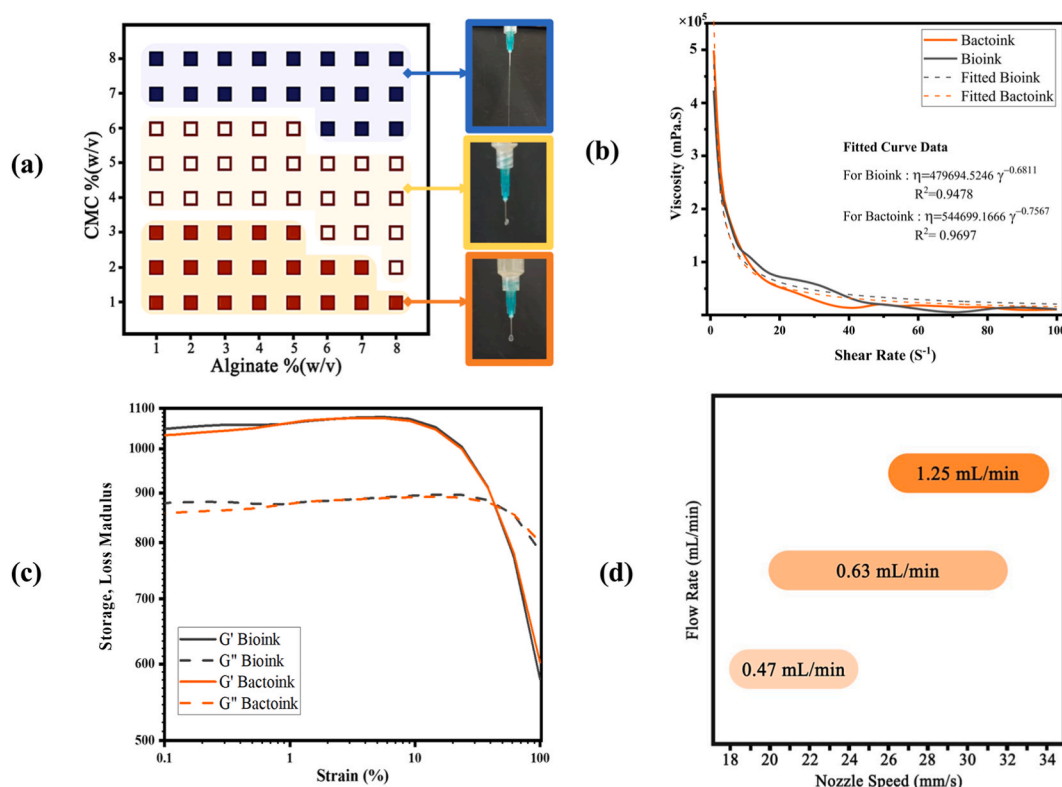


Fig. 3. Rheological Characterization of Bioinks: a) Analysis of viscosity; b) Storage (G') and loss (G'') moduli demonstrate the unique rheological properties of bacterial bioinks; c) Optimization of the alginate-to-CMC ratio to achieve optimal printability; d) Correlation between print speed and flow rate.

Where, η is viscosity, K is consistency index ($\text{Pa}\cdot\text{s}^n$), $\dot{\gamma}$ is shear rate, and n is flow index. Table 1 summarizes the fitted K and n and the corresponding R-squared values for bioink and bactoink.

Both bioink and bactoink exhibit strong shear-thinning behavior, as indicated by their flow indices (n) being significantly less than 1. The lower flow index of bactoink ($n = 0.243$) compared to bioink ($n = 0.319$) indicates that it undergoes more pronounced shear thinning, and its viscosity decreases more rapidly with increasing shear rate. For better comparison and more precise analysis of the flow behavior of bioink and bactoink, their viscosities at low (1 s^{-1}), moderate (10 s^{-1}), and high shear rates (100 s^{-1}) were calculated and summarized in Table 2.

At a shear rate of 1 s^{-1} (representing near-rest conditions), bactoink exhibits higher viscosity than bioink, suggesting a stronger internal microstructure. This enhanced structural integrity at rest is advantageous for bioprinting, as it promotes shape retention and fidelity immediately after extrusion. Conversely, under moderate to high shear rates (e.g., $10\text{--}100\text{ s}^{-1}$), bactoink demonstrates lower viscosity, facilitating smoother flow through the print nozzle and reducing extrusion pressure, thereby improving printability and process efficiency.

Based on Fig. 3c, the storage modulus (G') and loss modulus (G'') were found to be identical for both types of inks. Based on the storage modulus and the above discussion, the presence of bacteria slightly improved the ink shape retention or deformability after printing [42]. The loss factor (G''/G') being only slightly smaller than one suggests that the inks are not highly solid, proving their printability and ease of extrusion; however, their structural integrity has to be improved. Therefore, to provide further structural integrity, post crosslinking was conducted by immersion of the bio-printed constructs in CaCl_2 solution.

In the realm of bioprinting, achieving optimal resolution is intrinsically linked to the interplay between print speed and flow rate. The relationship between these two parameters is crucial, as both the extrusion rate of the syringe pump and the movement speed of the printhead significantly influence the printed line width [30]. Specifically, it has been established that enhancing printing resolution necessitates a careful adjustment of these parameters in tandem. For instance, as printhead movement speeds decrease, corresponding reductions in the syringe pump rates are required to maintain a consistent output without compromising quality. Furthermore, a suitable flow rate is essential for achieving a continuous and uniform extrusion during the printing process. Excessive flow rates can lead to over-extrusion, while inadequate rates may cause under-extrusion, resulting in discontinuities within the printed structure [43]. This gentle balance was explored in the current study by printing $20 \times 20\text{ mm}$ multilayer specimens (with a height of 20 mm) and visual inspection of geometric consistency, absence of filament buckling, splashing irregularities, deflections, and smooth filament deposition without gaps or overlaps. An optimal range of print speed to flow rate was determined to obtain the desired resolution for the bioink, as illustrated in Fig. 3d. Specifically, the optimal flow rates corresponding to the various nozzle speeds were determined to be 0.47 , 0.63 , and 1.25 mL/min . These values reflect the careful calibration necessary to achieve high-quality bioprinting with the specified gel, ensuring that the printed lines are both precise and continuous. Thus, the relationship between flow rate and print speed is not merely a matter of convenience but a fundamental aspect of bioprinting that directly impacts the fidelity and quality of the printed constructs. The findings underscore the importance of optimizing both parameters to enhance the overall printing resolution.

3.4. Characterization of 3D bioprinted constructs exposed to CO_2

3.4.1. SEM and elemental identification

Scanning Electron Microscopy (SEM) micrographs of 3D bioprinted constructs captured the morphological characteristics of CaCO_3 precipitates after in situ biomineral trapping of CO_2 (Fig. 4). The analysis revealed significant differences between the samples collected on day one (D1) and day two (D2). SEM images of the D1 sample exhibited dispersed precipitation of biominerals. In contrast, due to further biomineralization of the samples until the second day, an increase in the extent of precipitates was noticeable, with the calcium carbonates completely covering the surface. In this sample, the augmented precipitation was evident in both surface morphology and cross-sectional images.

SEM images also reveal the precipitate's polymorph type and size range. The SEM analysis indicates that the observed precipitates exhibit spherical or ovoid shapes, indicating the formation of vaterite [20]. The particle size of the calcium carbonate crystals ranged from approximately 2 to $5\text{ }\mu\text{m}$, in the same range as biocrystals observed in the previous studies [44]. Therefore, the SEM images point to the capability of 3D printed CO_2 absorbing living constructs to convert CO_2 to calcium carbonate crystals through sustained in situ carbon capture and biomineralization.

Energy Dispersive Spectroscopy (EDS) analysis corroborated the findings from the SEM images, revealing a significant increase in calcium carbonate production from D1 to D2, which can be attributed to continued bacterial activity. According to Fig. 5, the weight percentage of calcium element (an indicator of the precipitates) increased from 7% on Day 1 to 17% on Day 2, highlighting the effectiveness of the bacteria in facilitating mineralization over time.

Additionally, elemental mapping analysis (Fig. 5) was conducted to assess the spatial distribution of elements within the samples.

Table 1
The fitted consistency index (K) and flow index (n) for bioink and bactoink.

Sample	K ($\text{mPa}\cdot\text{s}^n$)	n	R^2
Bioink	476994.52	0.319	0.9478
Bactoink	544699.17	0.243	0.9697

Table 2
Calculated viscosities at low, moderate and high shear rates for bioink and bactoink.

Shear rate (s^{-1})	Bioink viscosity (mPa.s)	Bactoink viscosity (mPa.s)
1	476994.52	544699.17
10	99406.18	95379.87
100	20716.36	16701.55

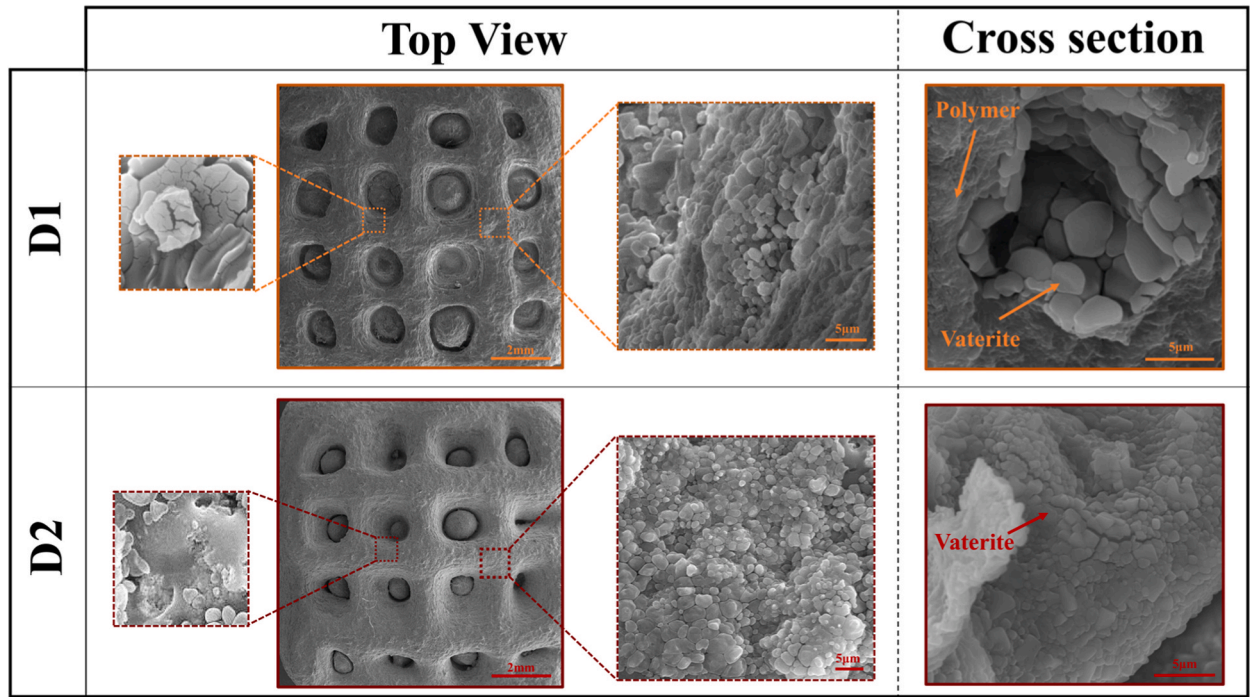


Fig. 4. SEM images of D1 and D2 samples (3D bioprinted constructs, exposed to CO₂ for one and two days, respectively). D2 exhibits the presence of a higher number of precipitates as compared to D1. Particle sizes range from 2 to 5 µm, typically in spherical or ovoid vaterite shapes. Both top and cross-sectional views are displayed.

The results, illustrated in Fig. 5, indicated a well-distributed presence of carbon, calcium, and oxygen throughout the samples. The elemental distribution exhibited a favorable scattering pattern, supporting the conclusion of sustained bacterial activity on D2 and the substantial increase in calcium carbonate content.

Collectively, the SEM and EDS analyses provided valuable insights into the morphological and compositional characteristics of calcium carbonate produced through bacterial processes. These findings highlight the potential of 3D bioprinting of bacteria to construct living CO₂ absorbing materials capable of in situ biomineralizing and mineralizing CO₂ trapping distributed across the structure.

3.4.2. XRD pattern

Fig. 6a illustrates the X-ray diffraction (XRD) patterns of the samples as a function of CO₂ exposure. The results indicate that during the initial 24 h (D1), vaterite peaks were hardly noticeable in a broad pattern attributed to the amorphous structure of the polymer matrix. Therefore, with a small amount of vaterite embedded within the polymer matrix, the peaks could not be distinctly identified. However, the characteristic peaks of the D2 samples clearly demonstrate the mineralization of calcium carbonate. As the incubation time increased, a pronounced peak at $2\theta = 32.45^\circ$ emerged, and the intensity ratio I_{D2}/I_{D1} was calculated to be 3.64. This peak corresponds to the (116) plane of hexagonal vaterite [45], aligning well with its known crystallographic characteristics (reference code: 96-901-3566). Since the XRD results clearly confirm the formation of the vaterite phase, the spherical structures observed in the SEM images, assumed to be vaterite, are validated. This correlation between crystallographic data and morphological features supports the phase identification. Moreover, the presence of these sharp peaks indicates the ongoing activity of bacteria in capturing CO₂. It is noteworthy that the formation of the vaterite structure occurs in a moderately alkaline medium and requires less energy [20]. To determine the average crystallite size of calcium carbonate produced by the bacteria, a value of $k = 0.94$ was selected, reflecting the spherical morphology of the particles. The full width at half maximum (FWHM) of the diffraction peaks was analyzed to calculate the average crystallite size using the Debye-Scherrer equation (Eq. (8)). The results indicated that the average crystallite size of vaterite

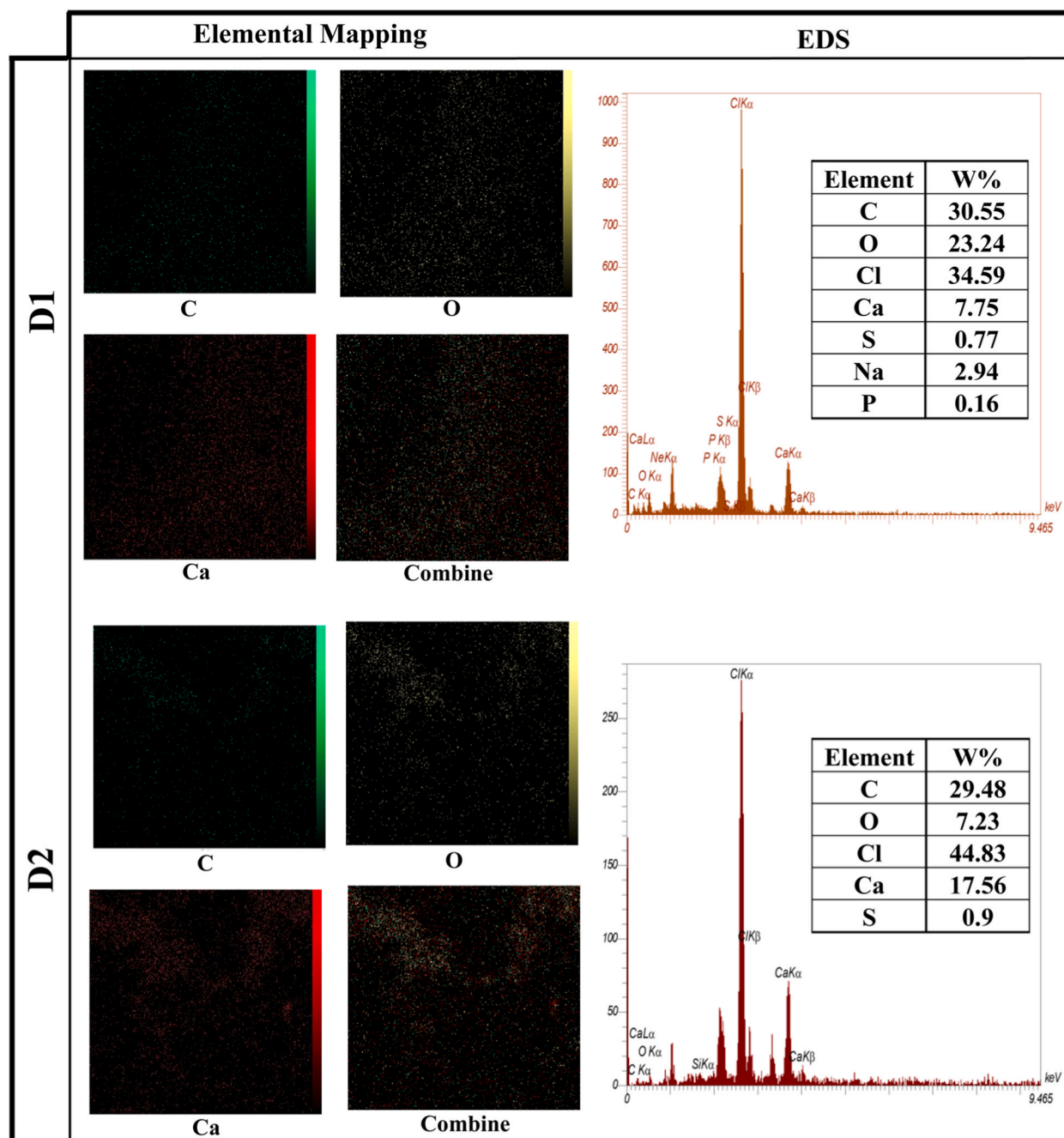


Fig. 5. EDS and Elemental mapping illustrate the significant increase in calcium carbonate production from D1 to D2, with weight percentages rising from 7% to 17%, indicating CO₂ biomineral trapping in the samples, as a result of bacterial activity.

was approximately 21 nm, as derived from the D2 X-ray diffraction data. This relatively small crystallite size can be attributed to the highly disordered nature of the vaterite structure [46]. Furthermore, a higher number of nucleation sites (i.e., bacterial cells) is another contributing factor behind the small crystallite sizes of vaterite obtained in this study [47]. In addition, the decrease in available calcium cations for biomineralization, as a result of CO₂ biomineral trapping, favors biomineralization in the form of vaterite polymorphs [28].

3.4.3. Raman spectroscopy

The presence of calcium carbonate was also confirmed through Raman spectroscopy analysis (Fig. 6b). Two samples were collected on the first and second days of the experiment. As the 3D biprinted constructs were exposed to sustained exposure to carbon dioxide,

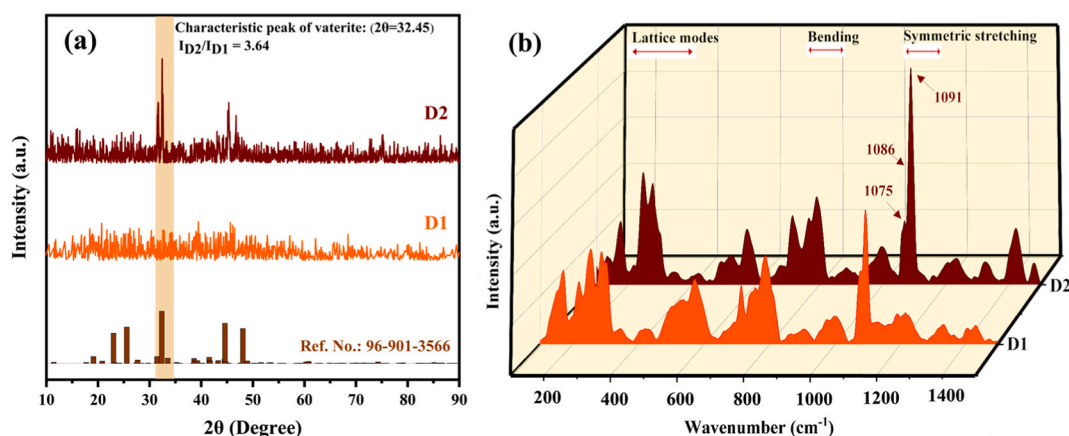


Fig. 6. a) XRD patterns of D1 and D2 reveal biomineralization in the form of vaterite polymorph. b) Raman spectroscopy confirms calcium carbonate presence in both samples, with day two showing stronger vaterite peaks (highly recognizable at 1075, 1086, 1091 cm^{-1}), indicative of bacterial CO_2 trapping via biomineralization.

the significant rise in the peaks observed on the second day reflects the biomineral tapping of CO_2 catalyzed through bacterial CA enzyme, continued on the second day. The calcium carbonate produced via this process exhibited characteristic peaks associated with vaterite, as evidenced by the presence of peaks in three primary regions: lattice mode, bending, and symmetric stretching [45]. Specifically, peaks at 1080 and 1090 cm^{-1} corresponded to the internal mode related to the symmetric stretching of the carbonate ion, while the in-plane bending mode was observed in the range of 700–800 cm^{-1} . Furthermore, peaks below 300 cm^{-1} , corresponding to translational and rotational lattice modes, were present in both samples, further supporting the identification of vaterite [46]. Therefore, the results of microfabric studies, including Raman Spectroscopy, XRD, and EDS, provide evidence of continued bacterially mediated biomineralization, cast in the form of vaterite polymorphs.

Semi-quantitative analysis of the D2 XRD pattern via Rietveld refinement indicates that the mineralized phase consists of **~93% vaterite and ~7% calcite**. This minor calcite fraction is consistent with the slight enhancement of the calcite-specific Raman band at $\sim 711 \text{ cm}^{-1}$ observed on Day 2 (Fig. 6b), likely resulting from partial transformation of metastable vaterite during incubation.

3.4.4. Thermogravimetric analysis (TGA)

Thermogravimetric analysis (TGA) was employed to investigate the thermal stability and decomposition characteristics of samples

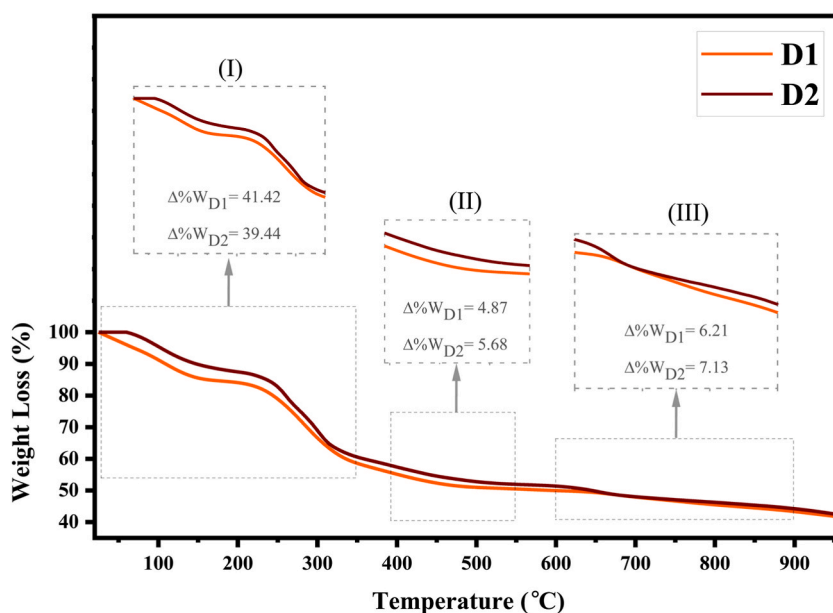


Fig. 7. TGA of samples D1 and D2 in a CO_2 -rich environment shows three weight loss stages: (I) moisture and thermal degradation of alginate and CMC; (II) vaterite to calcite transformation; (III) calcite decomposition into lime.

subjected to different incubation times in a CO₂-rich environment. Fig. 7 presents the TGA analysis results, which reveal significant insights into the structural transformations and weight loss patterns across different temperature ranges, which are crucial for characterizing the produced material.

The TGA results indicated distinct stages of weight loss corresponding to thermal stability. The first stage, occurring between 25 °C and 350 °C, is attributed to moisture loss and the thermal degradation of alginate and carboxymethyl cellulose (CMC). Notably, the weight loss up to 150 °C is primarily due to the evaporation of moisture [48], while complete degradation of CMC and alginate occurs between 250 °C and 350 °C [49], accounting for a total weight loss percentage of 41.42% and 39.44% for D1 and D2, respectively. The second stage of weight loss, identified between 395 °C and 540 °C [50], corresponds to the transformation of vaterite to calcite. This phase transition, characteristic of the material structure, exhibited weight changes of 4.87% for D1 and 5.68% for D2. The calculated carbonate content (obtained through equation (7)) indicates a weight percentage of 11.06% for D1 and 12.88% for D2, illustrating a 16.4% increase in vaterite content in the synthesized material. The third phase, occurring between 600 °C and 900 °C, signifies the decomposition of calcite into lime and carbon dioxide [51]. The weight loss measured during this stage was 14.1% for D1 and 16.19% for D2.

The results obtained from the thermogravimetric analysis (TGA) are aligned with our microfabric study using XRD analysis, indicating biological CO₂ sequestration in vaterite crystals. Results from TGA emphasize the influence of CO₂ exposure on the mineralization process. The analyzed samples (D1 and D2) were subjected to CO₂ in different durations, revealing a direct correlation between incubation time and carbonate content.

3.4.5. Compression test

Fig. 8 illustrates the outcome of the compression test on D1 and D2 samples. The results indicate that both samples exhibited similar mechanical behavior up to 10% strain, with average compressive strengths recorded at 3.4 MPa and 3.5 MPa, respectively. Upon further compression, cracks were initiated in the majority of the D1 samples at strains exceeding 16%. However, despite the emergence of cracks, the D1 samples maintained structural integrity up to 40% strain, where the sample could achieve a compressive strength of 17.2 MPa. It seems that the initiated crack has not quickly propagated within the sample, lowering its final strength. We attribute this phenomenon to the presence of biomineralized crystals stiffening the hydrogel matrix, thereby inhibiting the crack growth. The same fracture-resistance behavior and damage-resistance behavior have been seen in the natural biocomposites, for instance, spines of the sea urchin *Authoedaris Erassispina* (comprising biological nanocrystals surrounded by a matrix containing biomacromolecules) [13]. Therefore, in situ biominerals (formed through bacterial mineral trapping) embedded in a printed hydrogel matrix would not only increase the construct strength but also structural integrity.

After further biomineralization, D2 samples achieved an average compressive strength of 8.6 MPa at 40% strain. The strength decrease (induced by continued biomineralization) is potentially attributed to the formation of micropores due to biomineralization and bacterial activity. The continued growth of calcium carbonate crystals might create internal stresses. As more crystals form, they could push against the polymer chains of the hydrogel, potentially disrupting the matrix and creating micro-scale pores or weak interfaces between the crystals and the hydrogel [52]. By controlling the duration of CO₂ exposure, it may be possible to regulate the extent of crystal formation, thereby enabling tunable mechanical and structural properties in the 3D-printed hydrogel construct. A similar trend was reported in another study, where the maximum compressive strength reached 13.88 MPa after just 3 h of

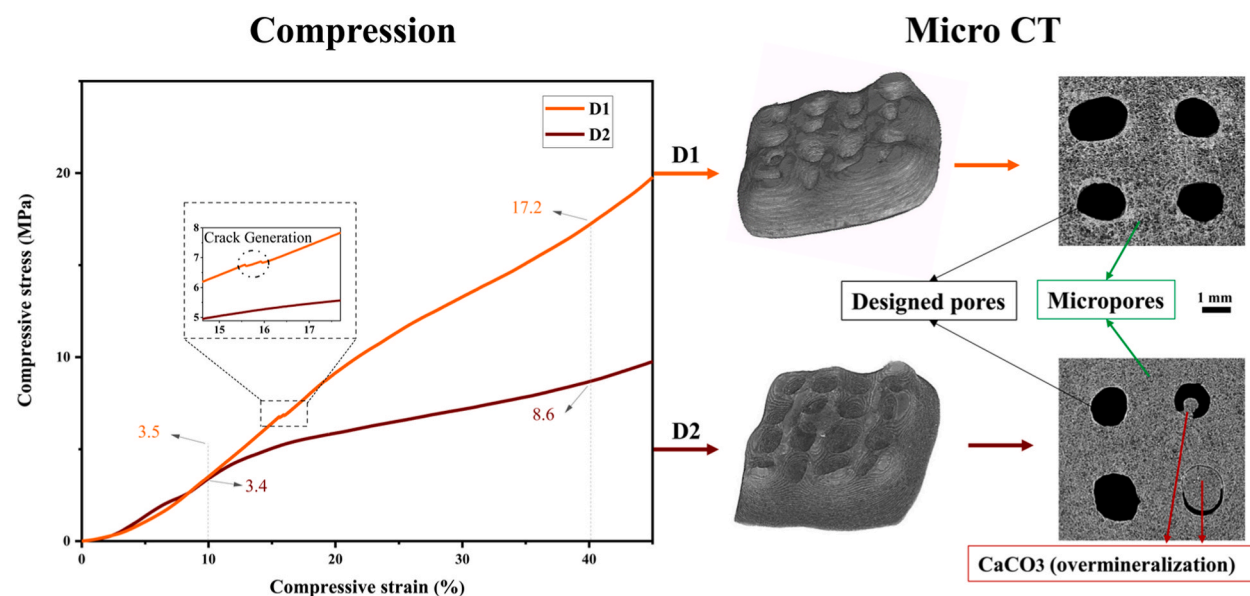


Fig. 8. Compressive strength results for D1 and D2 samples. Both samples showed similar behavior up to 10% strain. D1 and D2 samples achieved 17.2 MPa and 8.6 MPa strength, respectively, at 40% strain. Representative micro-CT images of the samples are also presented.

mineralization, followed by a 50% reduction to 6.96 MPa after 12 h [53]. As the crystals grew and aggregated, over-mineralization occurred, leading to the accumulation of CaCO_3 crystals on the fiber network. This aggregation weakened interfacial bonding and caused stress concentration, resulting in reduced elasticity, increased brittleness, and diminished mechanical properties. These findings further support the importance of controlling mineralization duration to preserve mechanical integrity.

In order to further inspect the potential increase in microporosity resulting from biomineralization and bacterial activity, X-ray μCT was conducted on the 3D bioprinted constructs after biomineralization. Quantitative analysis of the bioprinted constructs' matrix revealed microporosities of 37.83 % and 42.49% for D1 and D2 samples, respectively. As illustrated in Fig. 8, the engineered macropores became partially filled with biomineralized CaCO_3 in the D2 samples, leading to a measurable reduction in their diameters, a trend corroborated by SEM observations. The ongoing growth of calcium carbonate crystals likely generates internal stresses within the hydrogel matrix. As crystallization progresses, these expanding crystals may exert pressure on the surrounding polymer network, potentially disrupting the hydrogel structure and fostering the formation of microscale pores or weak interface between the mineral phase and the polymer matrix as also suggested in previous studies.

It is notable that control samples (prior to biomineralization) that were not exposed to CO_2 experienced considerable shrinkage and could not withstand the external load. Therefore, it can be deduced that the biologically mediated calcium carbonate precipitates act as a reinforcing element embedded into the polymeric matrix, thereby enhancing the structural integrity of these CO_2 absorbing living biocomposites as compared to the sole use of hydrogel. However, the level of biomineralization to achieve the required performance has to be controlled.

CO_2 biomineral trapping and in situ biomineralization considerably increased the strength of the printed constructs. When compared to available studies evaluating the compressive strength of 3D printed engineered living materials (such as those made of mycelium with a compressive strength of 60 kPa after 28 days [43]), the obtained strength in our study (up to 17.2 MPa in only one day) seems quite satisfactory. It is also noteworthy that the proposed method not only addresses CO_2 capture but also provides an energy-efficient eco-friendly technique for its biomineralization and utilization.

It is, therefore, recommended to stop the biomineralization at the desirable time to achieve the target strength and internal structure by deactivation of bacteria (i.e., over-mineralization could be prohibited by immersing the sample in ethanol to hinder the bacterial activity or through changing environmental conditions, such as temperature and pH). Furthermore, as a short CO_2 exposure period (i.e., one day) is adequate to achieve high strength, indicating a rapid curing time of 3D bioprinted constructs, the approach is deemed suitable for construction and industrial applications, where rapid curing is required.

3.4.6. Quantification of CO_2 sequestration capacity

Quantitative analysis revealed substantial biomineralized CaCO_3 precipitation within the bioprinted constructs. D1 samples exhibited a mean CaCO_3 content of 35.32 ± 3.59 wt%, and D2 samples showed a significantly higher mineralization level of 42.42 ± 2.16 wt%. This corresponds to a 20.1% relative increase in CaCO_3 content in D2 compared to D1, indicating a sustained mineralization. It also indicates that the mineralization has augmented over time (i.e., by further growth and activity of bacteria). This level of biomineralization not only confirms active microbial carbon sequestration but also suggests potential for structural reinforcement in bioprinted constructs, where CaCO_3 can serve as a natural biocement. However, the process has to be controlled to avoid over-mineralization, and achieve the desirable porosity and strength required for most industrial and biomedical applications. For this purpose, the activity of bacteria can be ceased at any time by the procedure stated in section 2.5.

CO_2 sequestration capacity is a key performance metric for CO_2 -absorbing materials, reflecting their ability to capture and mineralize carbon dioxide. In this study, the sequestration capacity was determined to be 5.46 mmol CO_2 per gram of matrix for D1 samples and 7.36 mmol CO_2 per gram of matrix for D2 samples (room temperature, 1atm absolute pressure), demonstrating a 34.8% enhancement in CO_2 uptake for the D2 formulation.

In light of the urgent need to mitigate climate change, numerous CO_2 sequestration strategies have been explored, but few balance high performance, scalability, and economic feasibility. Table 3 listed values CO_2 sequestration capacity of various methods. For instance, metal organic frameworks (MOFs) such as MOF-5 and MOF-177 achieve impressive capacities under high-pressure conditions (30 bar) [54]. However, their practical deployment is hindered by prohibitive synthesis costs and the energy-intensive infrastructure required to maintain elevated CO_2 pressures. In contrast, low-cost adsorbents like modified natural zeolites offer limited capacitances [55], far below the threshold needed for meaningful climate impact. Enzymatic approaches using purified carbonic anhydrase (CA) report exceptionally high activity [56], yet the high cost of enzyme extraction, poor thermal stability, and short shelf life render them impractical for large-scale use. Even among biological systems, the marine cyanobacterium *Synechococcus* achieves considerable capacities [57] but requires high cell densities and lacks structural utility.

Table 3
Comparison of CO_2 sequestration capacities of different methods.

Material	CO_2 Sequestration Capacity (mmol/g)	Ref.
MOF-5 (30 bar CO_2)	21.07	[54]
MOF-177 (30 bar CO_2)	32.5	[54]
Modified Zeolite	0.361	[55]
Purified Carbonic Anhydrase	2667.7	[56]
<i>Synechococcus</i> Bacteria	18	[57]
D1	5.46	This work.
D2	7.36	This work.

Having noted these challenges in available absorbents, our bioprinted system outperforms most practical alternatives on a per-matrix basis while using minimal bacterial biomass. With sequestration capacities of 5.46–7.36 mmol CO₂/g of composite matrix, our D2 formulation achieves >20 times higher performance than zeolites and approaches the efficiency of high-end MOFs, but under ambient conditions, without expensive materials or complex infrastructure. Moreover, unlike MOFs, enzymes, or planktonic cultures, our method integrates CO₂ sequestration directly into a functional material. The biomineralized CaCO₃ acts as an in situ strengthening agent, enhancing the mechanical integrity of the bioprinted construct. This dual functionality, carbon capture and structural reinforcement, combined with the use of low-cost *Sporosarcina pasteurii* at low inoculum density, positions our approach as a scalable, affordable, and value-added solution for sustainable carbon-negative manufacturing.

4. Conclusion

This study presents a novel approach for carbon dioxide capture through the integration of 3D bioprinting of living organisms with in situ biomineralization processes. A bioinspired biocomposite was synthesized using a hydrogel matrix of carboxymethyl cellulose and alginate, embedded with biomineralizing bacterial strains. The strain selected for its biomineralization capability was *Sporosarcina pasteurii*, facilitating the rapid conversion of CO₂ to bicarbonate ions and promoting calcium carbonate mineral formation within the printed constructs. The in situ biomineralization capacity of the 3D bioprinted constructs was effectively demonstrated following CO₂ exposure at 1 atm absolute pressure. Scanning electron microscopy revealed spherical morphology of precipitated calcium carbonates, while X-ray diffraction and Raman spectroscopy confirmed the presence of vaterite within the cell-laden constructs. Thermogravimetric analysis indicated sustained microbial viability and activity, evidenced by a 16.4% increase in calcium carbonate content by day 2. Mechanical testing showed that mineralized constructs achieved a compressive strength of up to 17.2 MPa on day 2, suggesting suitability for various industrial and biomedical applications. Furthermore, the sequestration capacities of bioprinted constructs reached 5.46 and 7.36 mmol CO₂/g for D1 and D2 specimens, respectively.

This study paves the way for integrating additive manufacturing and biocement capable of CO₂ capture, offering exciting opportunities for eco-friendly and sustainable construction as well as high-strength, functional living materials. These findings underscore the transformative potential of biocatalytic 3D printing to mitigate global carbon emissions and promote a sustainable future. Additionally, this approach highlights the potential of bioprinted, mineralized constructs as scaffolds for regenerative medicine. Importantly, this method directly utilizes CO₂, rather than relying on urea hydrolysis for biomineralization; consequently, the 3D bioprinted constructs are free of undesirable byproducts such as ammonium ions, which are produced in conventional (ureolytic) MICP approaches. Therefore, this method can be considered a viable alternative to existing techniques for cultural heritage repair that rely on urea hydrolysis.

5. Future outlook

This study focused on the feasibility of using bioprinted *S. pasteurii*-cell-laden constructs to capture CO₂ through mineral trapping, which showed promising results. The following suggestions can be made for future studies.

- As biomineralization can potentially help the 3D printed constructs to self-repair and heal, another aspect of the proposed approach can be self-healing. However, as the study was rather focused on the potential of in situ biomineralization inside these constructs after exposure to CO₂, for CO₂ capture and utilization, the self-healing aspects were not assessed. It can therefore be the subject of future studies.
- The implementation of the proposed technique for various applications such as art restoration, biomedical scaffold and tissue regeneration, repair of facades of historical buildings, as well as other civil and construction technological applications, has yet to be explored, with tailored experiments.
- Although *S. pasteurii* is generally regarded as non-pathogenic, any biomedical application (e.g., bone scaffolds) requires complete inactivation of residual bacteria post-mineralization to prevent immune responses. Future work should include rigorous assessment of biocompatibility, immunogenicity, and cytotoxicity of the sterilized constructs, as well as validation of mechanical stability under physiological conditions.
- Inspired by natural biocomposites such as sea urchin spine composition (i.e., a biocomposite consisting of an organic matrix and biominerals), this study considered in-situ biomineralization through microbial cells embedded in a biopolymer matrix for CO₂ capture and material stiffening. However, there are several choices of biopolymers, which can be explored in future studies, providing various strength levels, structures, and printability for diverse applications and functions.
- This study utilized *S. pasteurii* strains (without any genetic modifications). Carbonic anhydrase enzyme, an essential part of the proposed method, is encoded in the genome of *S. pasteurii* (i.e., γ -CA and β -CA with WP_115362601 and WP_115363014 NCBI Reference Sequence numbers, respectively). Overexpression of the respective genes and upregulation of the CA enzyme in genetically engineered strains can increase biomineralization and consequently CO₂ capture. Integrating 3D bioprinting with bespoke genetically engineered strains unlocks unprecedented potential for industrial scale 3D printed living materials for CO₂ capture.
- The long-term durability of the 3D-printed bactoink constructs is highly dependent on their intended application environment. In industrial or construction settings, environmental stressors such as cyclic humidity, UV exposure, temperature fluctuations, and acidic rains may induce polymer swelling, photodegradation, or partial dissolution of biomineralized CaCO₃. These factors necessitate systematic accelerated aging studies. In contrast, biomedical applications require temporal rather than indefinite

stability, as the constructs must maintain mechanical integrity during early tissue integration and subsequently undergo controlled resorption. Although similar MICP hydrogel systems have already demonstrated sufficient stability for indoor cultural heritage restoration [20] and biopolymer-biocementation treatments have shown promising durability [58], the long term durability of the proposed constructs still needs dedicated studies. To enhance environmental robustness, future works should explore UV-shielding or reinforcing nano-additives and evaluate construct performance under cyclic humidity, temperature, and acidic conditions.

- Investigation of the CO₂ sequestration performance of the bactoink constructs under more realistic atmospheric conditions, including ambient CO₂ concentrations, controlled airflow, and variable relative humidity is another gap between laboratory and industrial application which highlights studies that provide insight into gas-transport dynamics, hydration effects, and mass-transfer limitations that govern mineralization in open systems. Extending testing beyond sealed CO₂ chambers, will establish the operational feasibility and environmental applicability of this low-pressure, biologically driven CO₂ capture approach.
- The potential use of bactoink in off-Earth construction warrants systematic investigation under the extreme conditions' characteristic of Martian and lunar environments. The intrinsic stability of the biocemented CaCO₃ phase provides promising thermal and radiation tolerance, while the alginate/CMC hydrogel offers a printable and protective microenvironment during early mineralization. The primary function of bacteria is to facilitate mineral precipitation. Their long-term survival is not necessarily required for structural integrity once the biocement has formed. Future research should evaluate strategies to enhance biological robustness, including co-culturing biomineralizing bacteria with radiation-resistant extremophiles (such as *Deinococcus radiodurans* and *Chroococcidiopsis*) as well as encapsulation approaches (such as carbon-fiber encapsulation [59]) that shield cells during bio-mineralization. Additional protective strategies-such as temporary shielding during printing or incorporation of UV-blocking additives into the hydrogel could also be considered. In parallel, leveraging in-situ resource utilization using Martian atmospheric CO₂ and local regolith as carbon and aggregate sources could substantially reduce material transport requirements. Overall, advancing bactoink toward extraterrestrial applications will require dedicated studies on printability, microbial survival, and composite performance under low pressure, high radiation, and severe temperature fluctuations.

6. A road map to technology scale up: envisaging hurdles and potential solutions

The proposed methodology has shown promising results in terms of compressive strength and CO₂ capture potential. However, in order to scale up this technology, future advancements are required to facilitate the use of this technology at industrial scale. For this purpose, the following suggestions are set forth.

- a. **Mass transfer and CO₂ deliverability enhancement:** To increase the amount of CO₂ mineralized by the embedded bacteria in bioconstructs, the optimum thickness of bioconstructs has to be obtained so that CO₂ diffusion is enhanced with minimum impact on the target mechanical strength. For industrial scale printing, a sequence of printing, CO₂ exposure, in situ mineralization, and printing next layers can also be considered. Modular construction using the printed sections would be another suggestion. Sacrificial or tailored channels for enhanced CO₂ diffusion would also facilitate mass transfer into the constructs to reach embedded bacteria.
- b. **Enhancing spatial distribution of bacteria:** Spatial patterning and controlling the location of bacteria via bioprinting can increase uniformity of the spatial bacterial distribution across the printed sample and sections, leading to more controlled mechanical properties. For special cases, nonuniform controlled distribution of bacteria for constructions with different strength sections without changing material can also be considered.
- c. **Controlling biomineralization speed and reagents availability:** Moreover, instead of directly utilizing bacteria in bactoink, the potential of sustained and controlled release of chemicals required for biomineralization using encapsulated bacteria and cementation reagents added to bactoink could be inspected (i.e., embedment of cementitious reagents in an engineered porous medium, such as Metal-organic frameworks (MOFs) to control biomineralization speed). In addition, MOFs can absorb CO₂ on their surface due to their unique tunable pore structures and act as another source of CO₂ and facilitate CO₂ supply for bacterial mineralization. Hurdles such as longer-term nutrient supply and culture stability can be potentially overcome by encapsulating bacteria, nutrients, and cementation reagents before their inclusion in the bactoink. Embedment of lyophilized bacteria help a longer, sustained, and enhanced biomineralization with further control on biochemical reactions and bacterial growth [60]. Controlled growth and deposition of minerals would eventually help to achieve controlled porosity and strength. This advancement would better inform the design of mechanically and biologically adaptive bactoinks for applications such as designing scaffolds.
- d. **Process integration and throughput:** Upscaling and industrialization of the proposed technology requires integration with continuous-flow bioreactors or modular systems capable of handling industrial gas streams. Parallelization and automation of printing, as seen in tissue engineering, could be targeted for this purpose. The use of multi-nozzle printing facilities and AI-driven print path optimization for maximal bacterial viability and CO₂ contact would increase throughput. Direct coupling with flue gas or direct air capture (DAC) units is technically feasible but requires system-level optimization to manage gas composition, humidity, and pressure, challenges acknowledged in broader CCS scaling efforts.
- e. **Augmenting structural integrity and mechanical strength:** Development of hybrid materials (e.g., polymer-ceramic composites) to withstand mineral expansion could also be foreseen as another future advancement (i.e., where over-biomineralization is concerned). In addition, long-term durability of constructs under UV radiation, acid rain, cyclic humidity and temperature fluctuations is of prominence.

CRedit authorship contribution statement

Hamed Aghili: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Ehsan Khodayari:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Sajjad Deylaghian:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Ehsan Nikooee:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Conceptualization. **Elahe Parvizmonfared:** Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Sirus Javadpour:** Writing – review & editing, Validation, Supervision, Resources, Formal analysis. **Mojtaba Ansari:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Mohammad Reza Jahanmard:** Writing – original draft, Methodology, Investigation, Formal analysis.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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